

# MATH 345 Skeleton Notes

## Unit 6: Constraints and SVD

### Section 6.5: Low-rank approximation and compression

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*Application question.* How can a matrix be approximated by keeping only its strongest directions?

The SVD writes a matrix map as a sum of directional pieces. If

$$A = U\Sigma V^T$$

has rank  $r$ , then

$$A = \sum_{i=1}^r \sigma_i \mathbf{u}_i \mathbf{v}_i^T.$$

The matrix

$$\sigma_i \mathbf{u}_i \mathbf{v}_i^T$$

has rank one. It measures the input coordinate in the direction  $\mathbf{v}_i$ , stretches by  $\sigma_i$ , and sends the result in the output direction  $\mathbf{u}_i$ .

**Definition: Rank- $k$  reconstruction** Let

$$A = U\Sigma V^T$$

be an SVD of a rank- $r$  matrix  $A$ . For  $1 \leq k \leq r$ , the rank- $k$  reconstruction of  $A$  is

$$A_k = U_k \Sigma_k V_k^T = \sum_{i=1}^k \sigma_i \mathbf{u}_i \mathbf{v}_i^T,$$

where  $U_k$  contains the first  $k$  columns of  $U$ ,  $\Sigma_k$  contains the first  $k$  singular values, and  $V_k$  contains the first  $k$  columns of  $V$ .

The matrix  $A_k$  keeps the  $k$  largest singular directions and discards the rest. Its rank is at most  $k$ .

### Activity: Rank-one reconstruction of a diagonal stretch

Return to the diagonal stretch from Section 6.3:

$$A = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}.$$

For this matrix, the right and left singular vectors are

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix},$$

and the singular values are

$$\sigma_1 = 3, \quad \sigma_2 = 1.$$

1. Write  $A$  as

$$3\mathbf{e}_1\mathbf{e}_1^T + \mathbf{e}_2\mathbf{e}_2^T.$$

2. Compute the rank-one reconstruction

$$A_1 = 3\mathbf{e}_1\mathbf{e}_1^T.$$

3. For

$$\mathbf{x} = \begin{bmatrix} a \\ b \end{bmatrix},$$

compute  $A\mathbf{x}$  and  $A_1\mathbf{x}$ .

4. Which input direction is discarded by  $A_1$ ?
5. What is  $\text{rank}(A_1)$ ?

**Note: Energy retained** A common way to summarize how much of a matrix has been kept is to use squared singular values. The energy retained by the first  $k$  singular values is

$$\frac{\sigma_1^2 + \cdots + \sigma_k^2}{\sigma_1^2 + \cdots + \sigma_r^2}.$$

This number is between 0 and 1. Larger values mean that the rank- $k$  reconstruction keeps more of the matrix's dominant directions.

**Activity: Choosing  $k$  from singular values**

Suppose the nonzero singular values of a matrix are

10, 4, 1, 0.5, 0.1.

1. Compute the total squared singular-value energy.
  2. Compute the energy retained by  $k = 1$ .
  3. Compute the energy retained by  $k = 2$ .
  4. What is the smallest  $k$  that retains at least 90% of the energy?
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**Note: Rank- $k$  reconstruction in NumPy** In NumPy, the singular values are returned as a one-dimensional array  $s$ . The diagonal matrix  $\Sigma_k$  is built from  $s[:k]$ .

```
U, s, Vt = np.linalg.svd(A, full_matrices=False)
Ak = U[:, :k] @ np.diag(s[:k]) @ Vt[:k, :]
```

The matrix  $U[:, :k]$  keeps the first  $k$  left singular vectors. The matrix  $Vt[:k, :]$  keeps the first  $k$  rows of  $V^T$ , which correspond to the first  $k$  right singular vectors. The product  $A_k$  is the rank- $k$  reconstruction. For a quick reference on SVD commands, diagonal matrices, slices, and matrix multiplication, see the programming appendix sections NumPy Linear Algebra Commands, Constructing Common Arrays and Design Matrices, Indexing and Slicing, and Elementwise Arithmetic Versus Linear Algebra.

### Activity: Reading rank- $k$ reconstruction code

The following code computes a rank-one reconstruction of the diagonal stretch.

```
import numpy as np

A = np.array([[3., 0.],
              [0., 1.]])

U, s, Vt = np.linalg.svd(A, full_matrices=False)

k = 1
Ak = U[:, :k] @ np.diag(s[:k]) @ Vt[:k, :]

s, Ak
```

Output:

```
(array([3., 1.]),
 array([[3., 0.],
```

```
[0., 0.]]))
```

1. What are the singular values of  $A$ ?
2. Which singular value is kept when  $k = 1$ ?
3. What is the matrix  $A_1$ ?
4. Which input direction does  $A_1$  discard?
5. Why is  $Vt[:, :k]$  used instead of  $Vt[:, :k]$ ?

### Activity: Redundant features revisited by singular values

Return to the redundant-feature matrix from Unit 2:

$$X = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 0 & 1 & 1 \\ 1 & -1 & 0 \end{bmatrix}.$$

The third column is the sum of the first two columns.

The following code computes the singular values and the last row of  $V^T$ .

```
import numpy as np

np.set_printoptions(precision=3, suppress=True)
```

```
X = np.array([[1., 2., 3.],
              [2., 4., 6.],
              [0., 1., 1.],
              [1., -1., 0.]])

U, s, Vt = np.linalg.svd(X, full_matrices=False)

s, Vt[-1, :]
```

Output:

```
(array([8.478, 1.459, 0.   ]),
 array([-0.577, -0.577, 0.577]))
```

1. What does the zero singular value say about the rank of  $X$ ?
2. What null-space direction does the last row of  $V^T$  point along?
3. How does this match the Unit 2 relation among the columns?
4. If  $X\mathbf{c} = \mathbf{y}$ , what happens to  $X(\mathbf{c} + t\mathbf{z})$ , where

$$\mathbf{z} = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} ?$$

**Note: Low-rank updates** A low-rank update changes a large matrix by adding a product

$$BC,$$

where

$$B \in \mathbb{R}^{m \times r}, \quad C \in \mathbb{R}^{r \times n},$$

and  $r$  is small. Since

$$\text{rank}(BC) \leq r,$$

the update  $BC$  changes the matrix through at most  $r$  independent directions.

This is useful when a full  $m \times n$  matrix would be expensive to store or change. The matrix  $BC$  can be stored using

$$mr + rn$$

numbers instead of  $mn$  numbers.

In model-adaptation settings,  $W$  may be a large existing weight matrix and  $BC$  may represent a small trainable update. The linear-algebra point is independent of that application: the update  $BC$  has rank at most  $r$  and can be stored using many fewer numbers than a full  $m \times n$  matrix.

### Activity: Rank of an update

Let

$$W \in \mathbb{R}^{1000 \times 800}, \quad B \in \mathbb{R}^{1000 \times 4}, \quad C \in \mathbb{R}^{4 \times 800}.$$

Define

$$W_{\text{new}} = W + BC.$$

1. What is the shape of  $BC$ ?
2. What is the largest possible rank of  $BC$ ?

3. How many numbers are needed to store  $W$ ?
4. How many numbers are needed to store  $B$  and  $C$ ?
5. Why is  $BC$  called a low-rank update?

**Warning: Low-rank update versus low-rank matrix** A low-rank update does not mean that the whole matrix

$$W_{\text{new}} = W + BC$$

has low rank. The update  $BC$  has rank at most  $r$ . The matrix  $W + BC$  may still have large rank.